

E9 205 Machine Learning for Signal Processing

Considerations in Deep NN learning

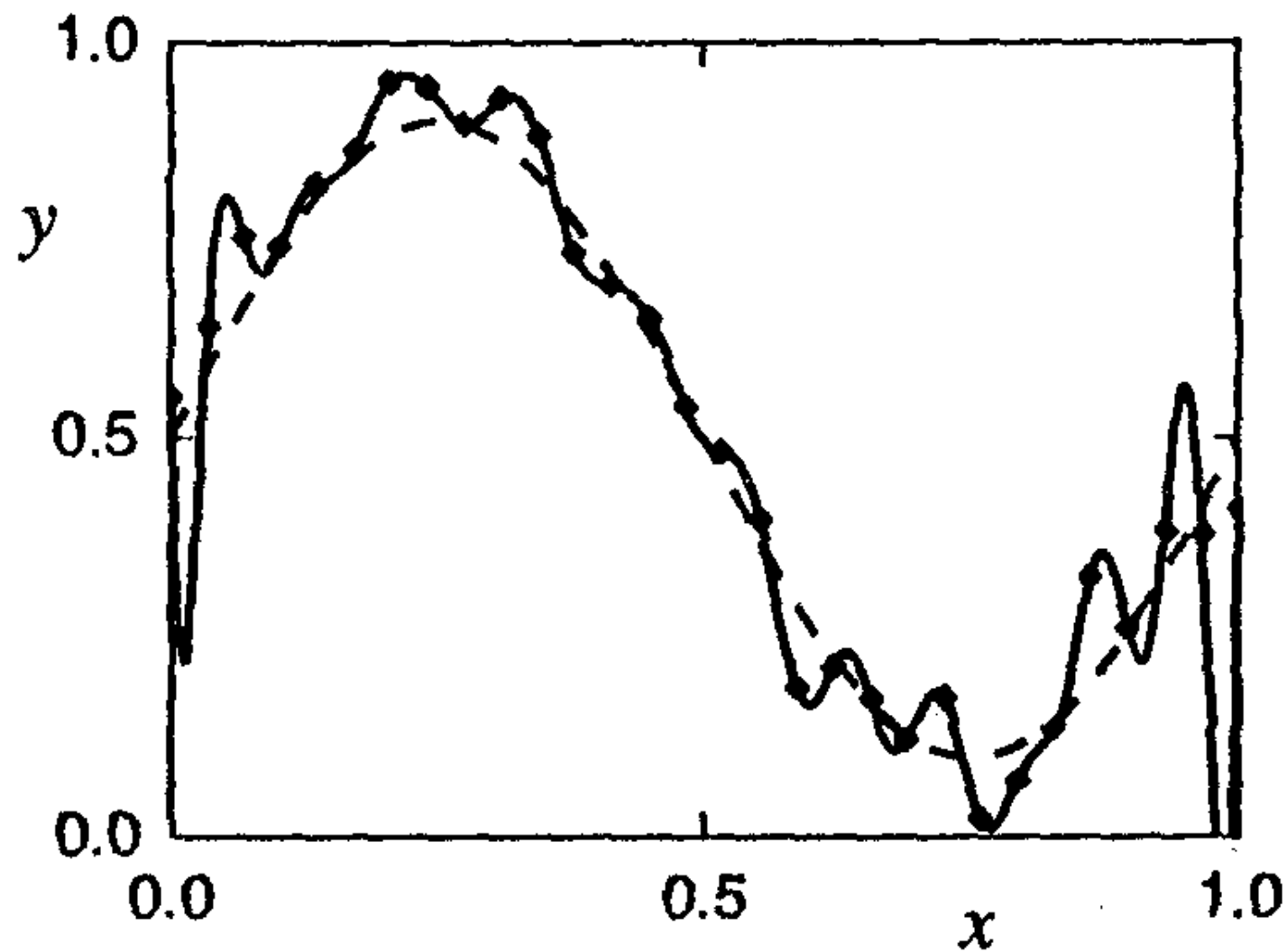
15-11-2017

Bias Variance Trade Off and Overfitting

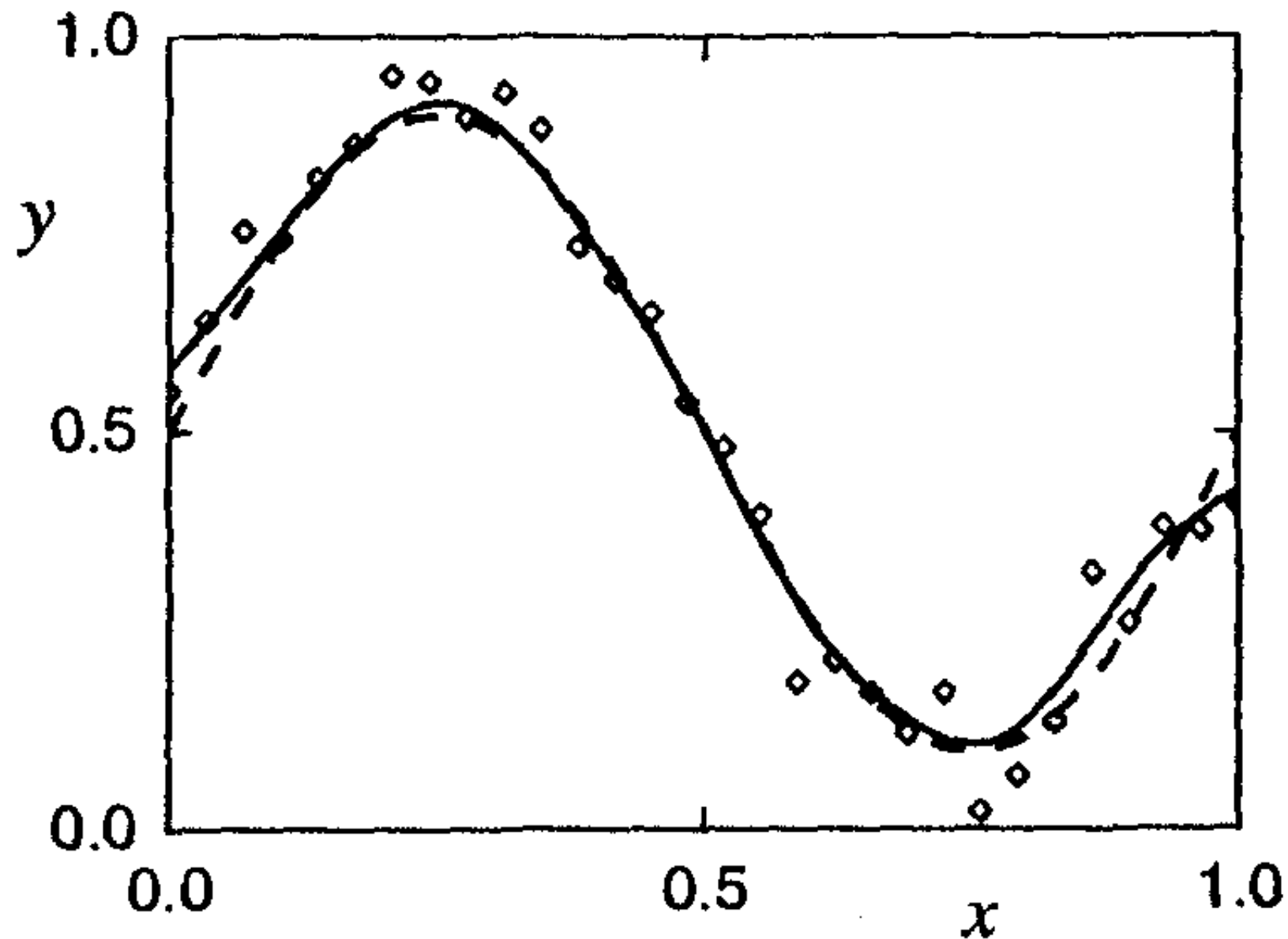
$$(\text{bias})^2 = \frac{1}{2} \int \{\mathcal{E}_D[y(\mathbf{x})] - \langle t|\mathbf{x} \rangle\}^2 p(\mathbf{x}) d\mathbf{x}$$

$$\text{variance} = \frac{1}{2} \int \mathcal{E}_D[\{y(\mathbf{x}) - \mathcal{E}_D[y(\mathbf{x})]\}^2] p(\mathbf{x}) d\mathbf{x}.$$

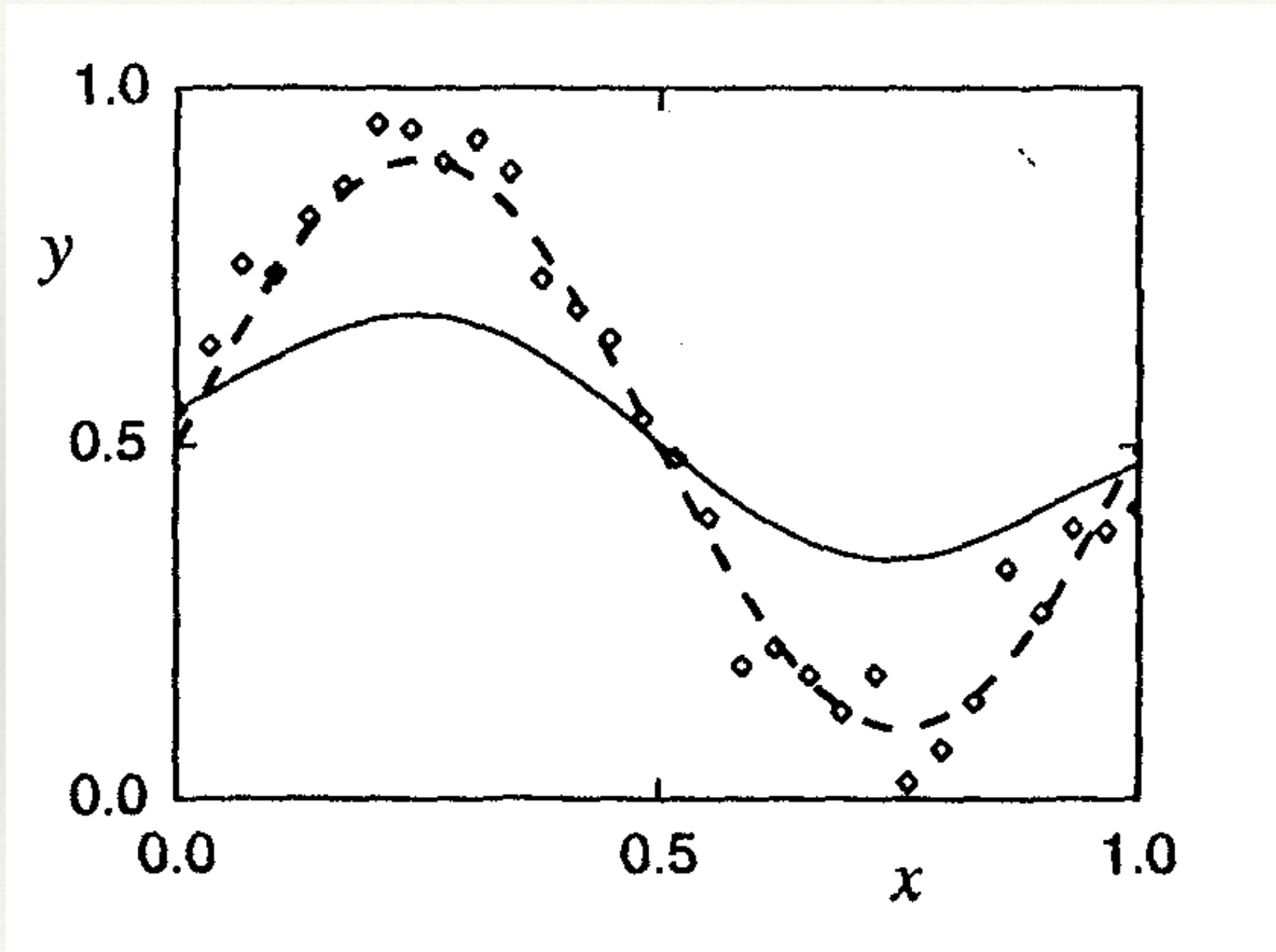
Bias Variance Trade Off and Overfitting



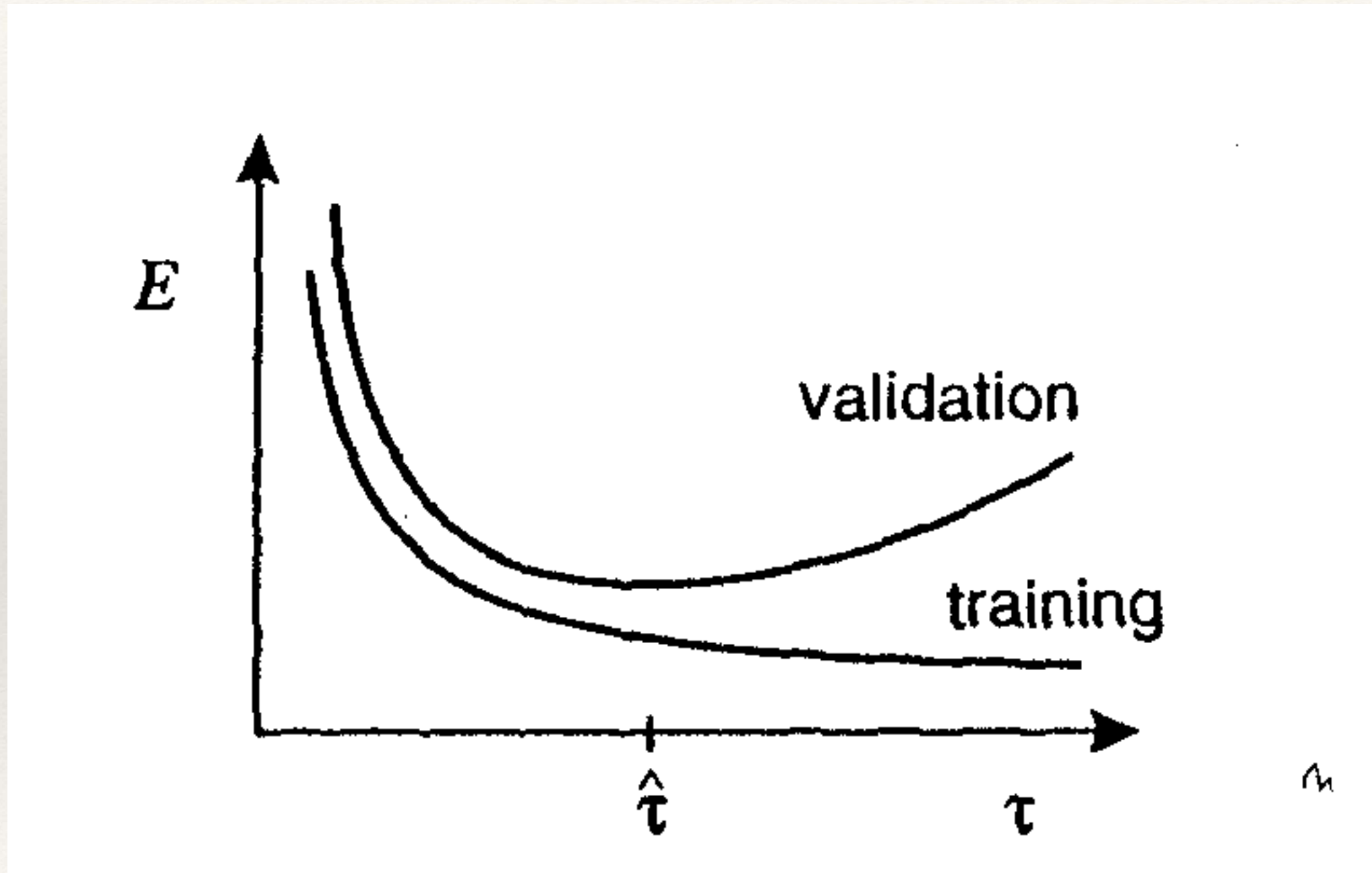
Weight Decay Regularization



Weight Decay Regularization



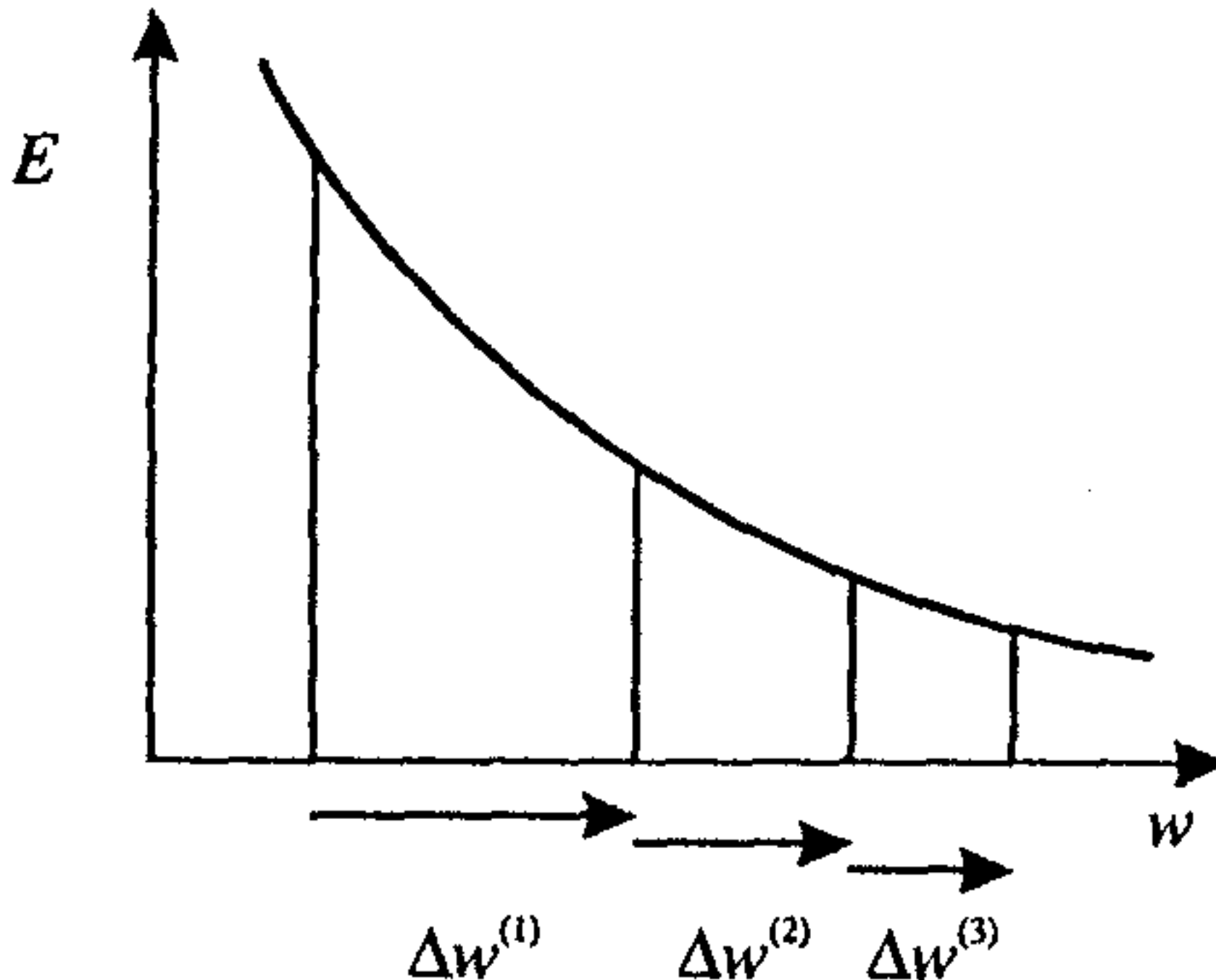
Early Stopping



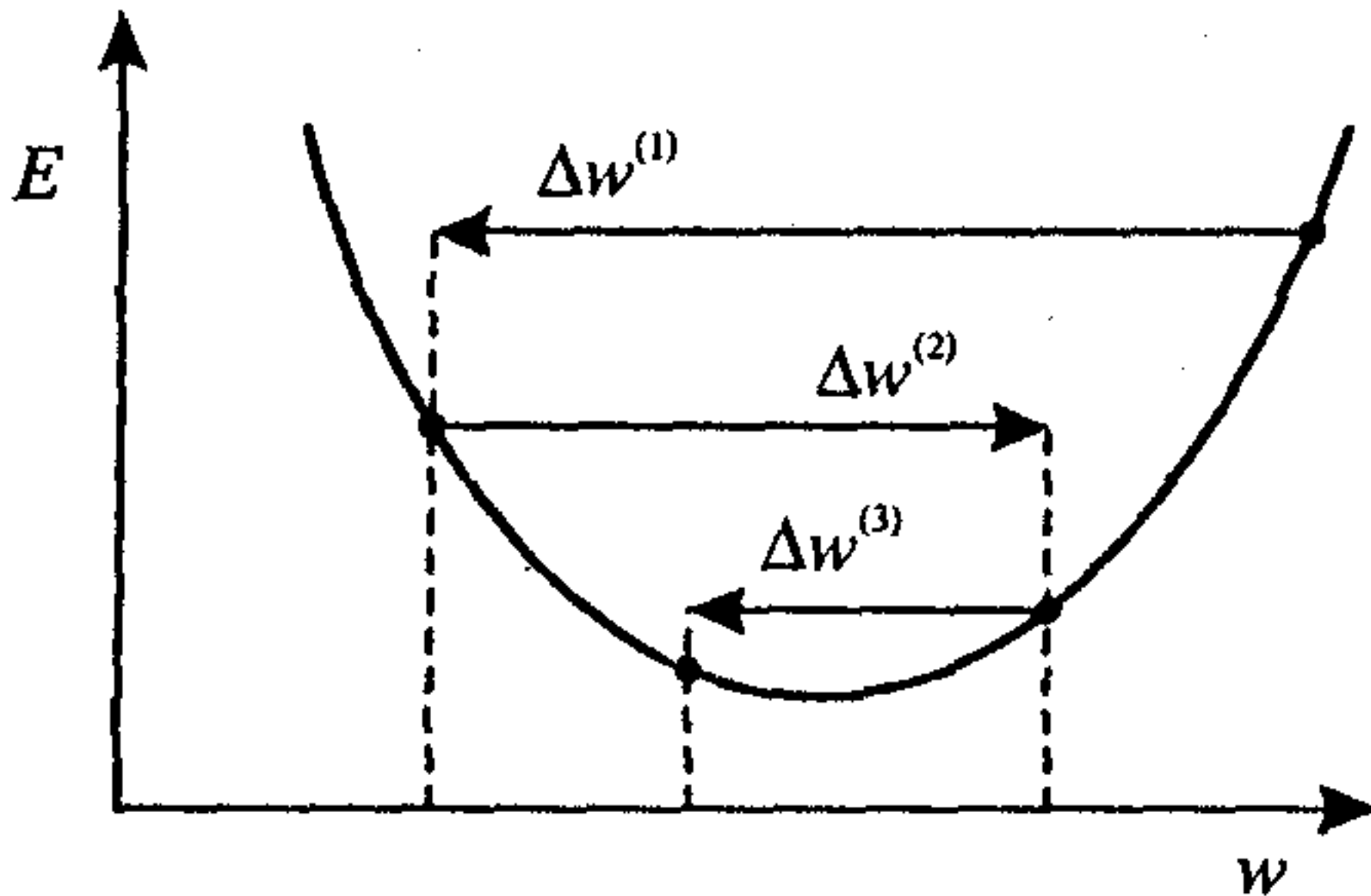
Attempts to Improve Learning

- ❖ Increasing the amount of training data
- ❖ Regularization (eg. weight decay)
- ❖ Early stopping using cross validation
- ❖ Training data with noise.
- ❖ Committees of Neural Networks

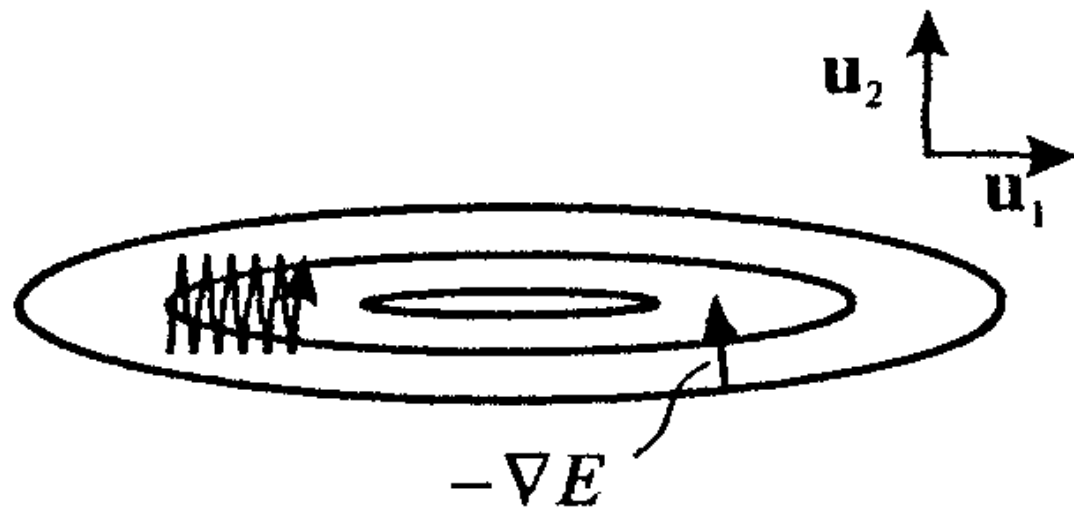
Momentum



Momentum

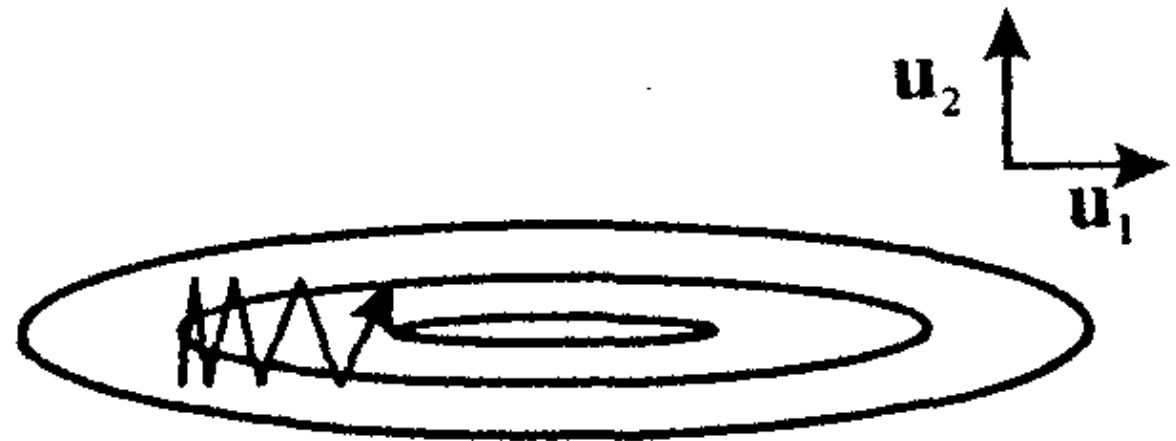


Momentum



Without Momentum

With Momentum

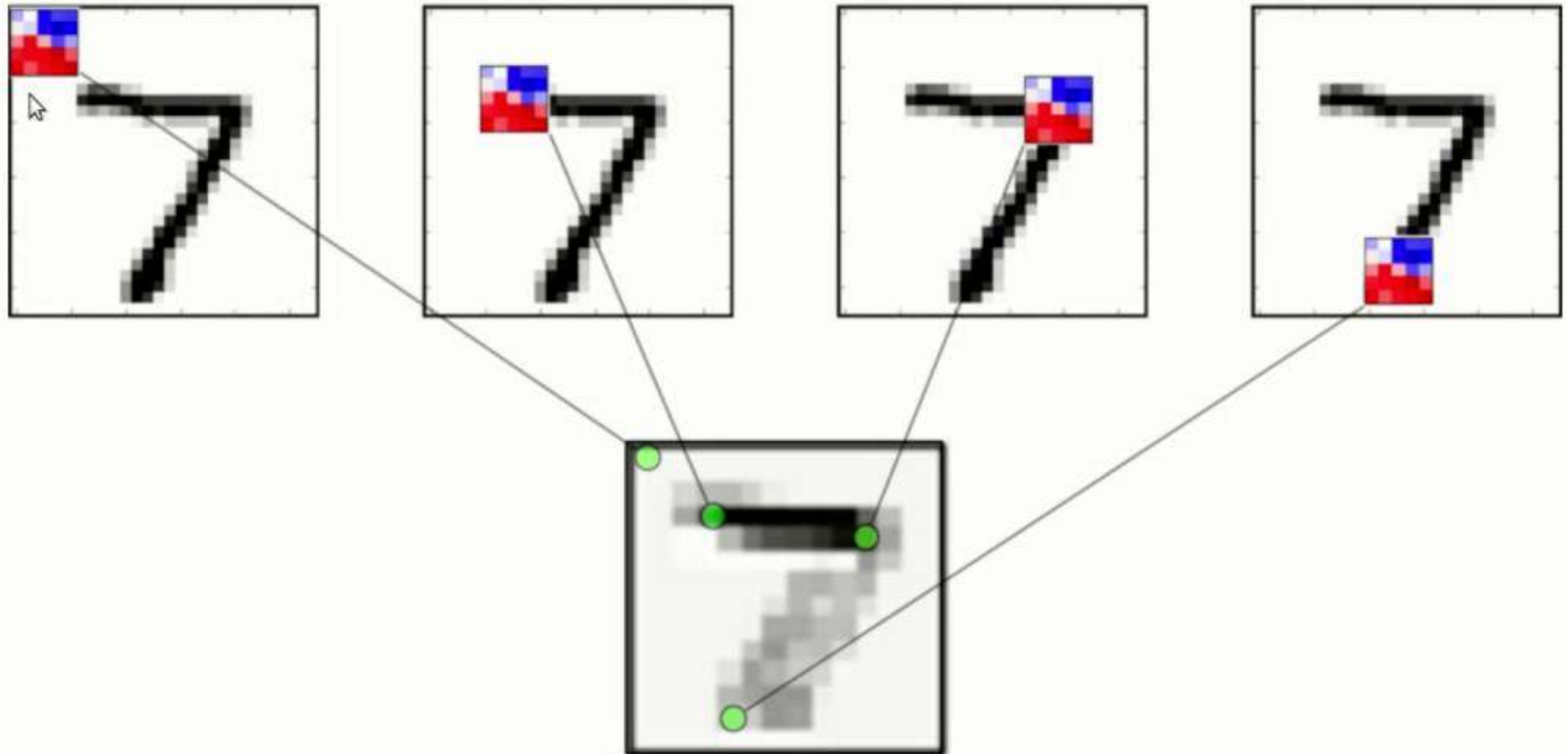


Convolution Operation (Images)

File Edit View Insert Cell Kernel Help Python 3

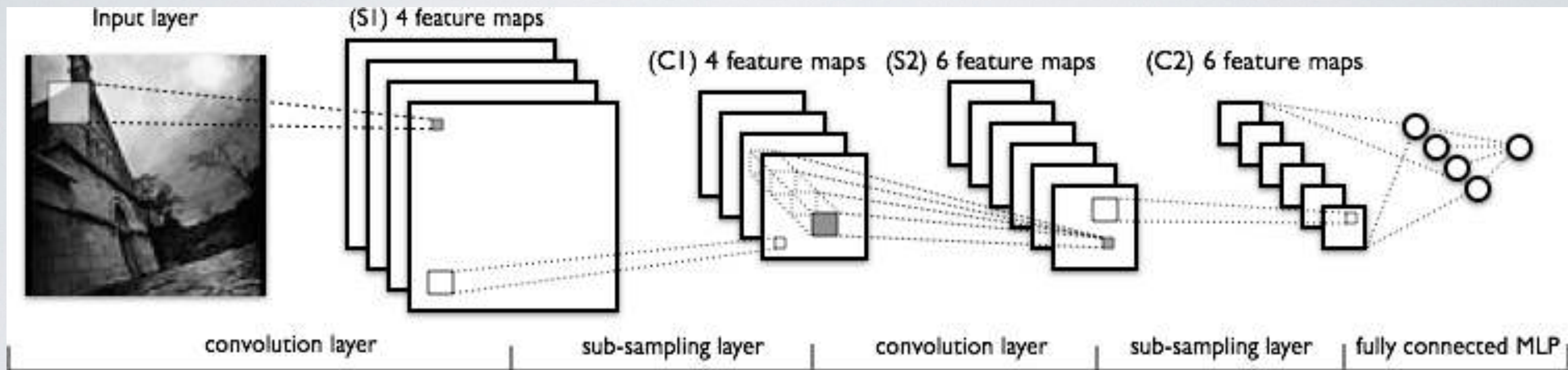
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Image('images/02_convolution.png')
```

Out[2]: Input Image with Filter Overlaid (4 copies for clarity)



Result of Convolution

Convolutional Networks



- Multiple levels of filtering and subsampling operations.
- Feature maps are generated at every layer.

CNNs for Speech

